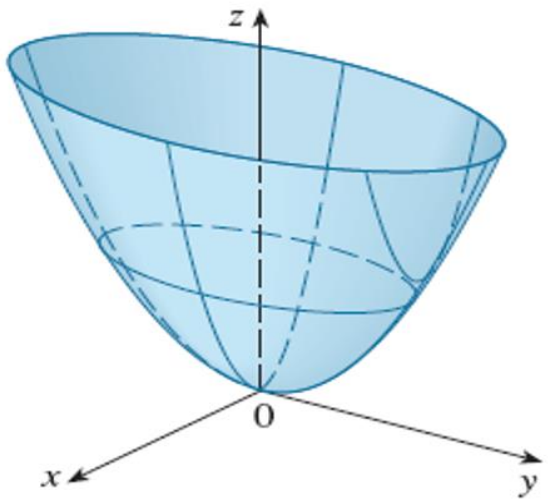
Lớp giải tích 2 online

HCMUT – CNCP

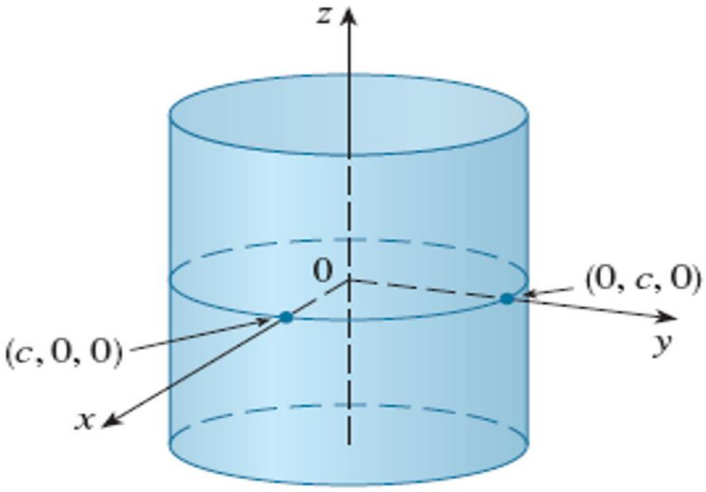
Mặt bậc hai

1/ Mặt paraboloid elliptic $z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$

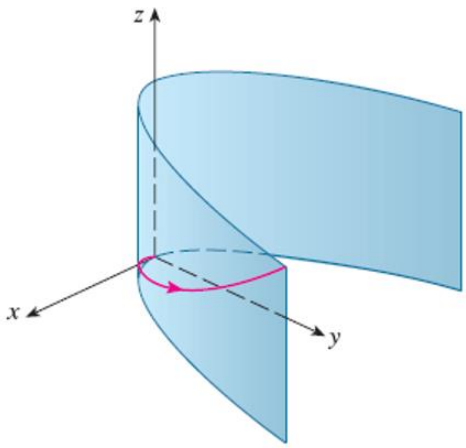


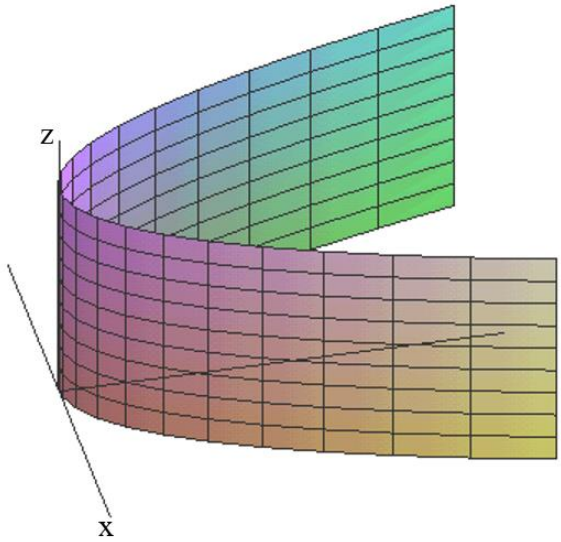
2/ Mặt trụ: trong phương trình thiếu hoặc x, hoặc y, hoặc z.

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

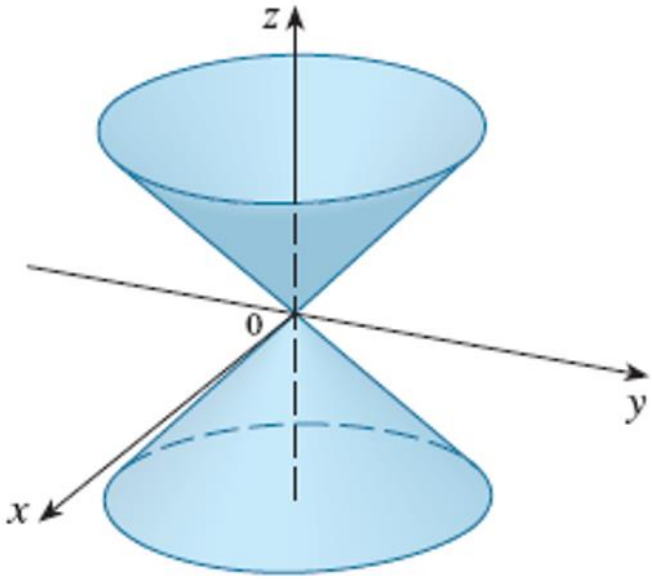


Mặt trụ $y = x^2$



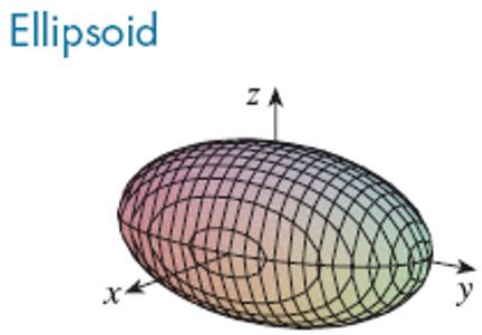


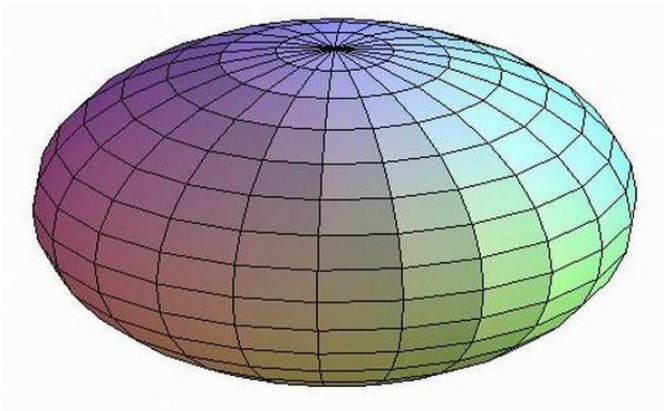
3/ Mặt nón hai phía $z^2 = \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2}$



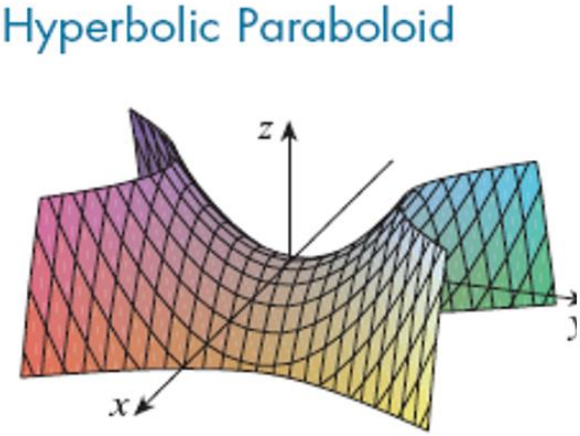
Mặt nón một phía $z = \sqrt{\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2}}$

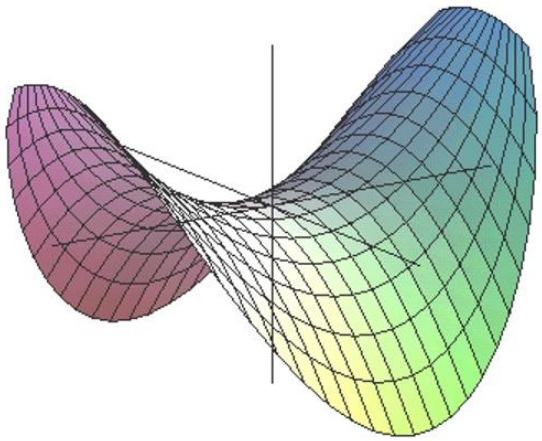
4/ Mặt ellipsoid $$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$$



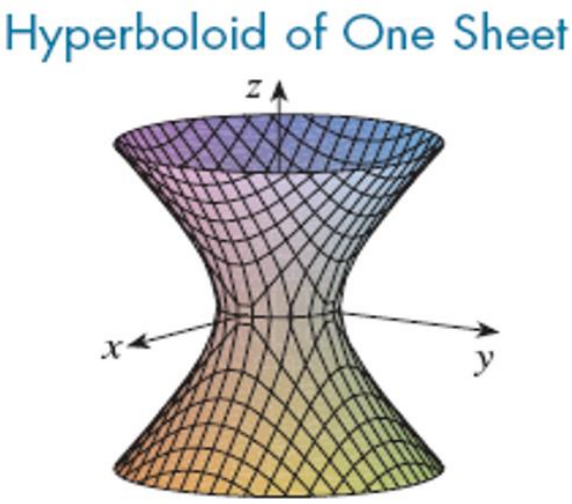


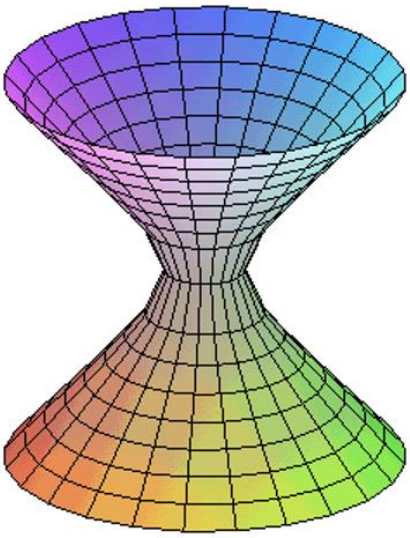
5/ Mặt Paraboloid hyperbolic $$z=\frac{x^2}{a^2}-\frac{y^2}{b^2}$$



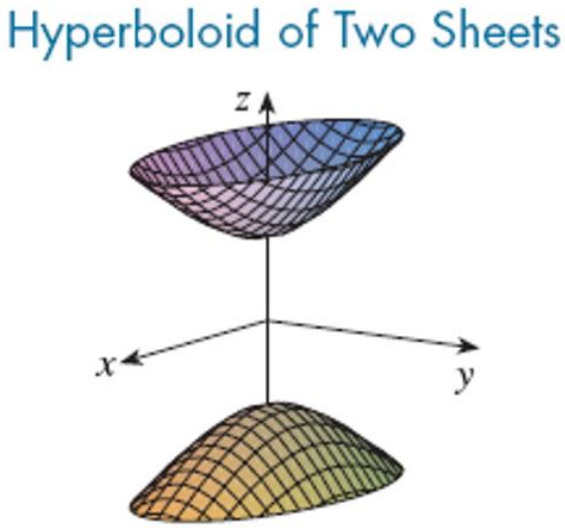


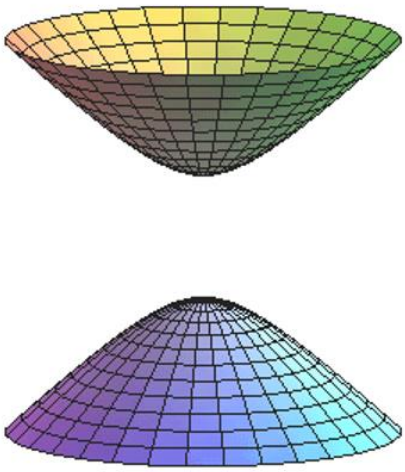
6/ Mặt Hyperboloid 1 tầng $\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=1$



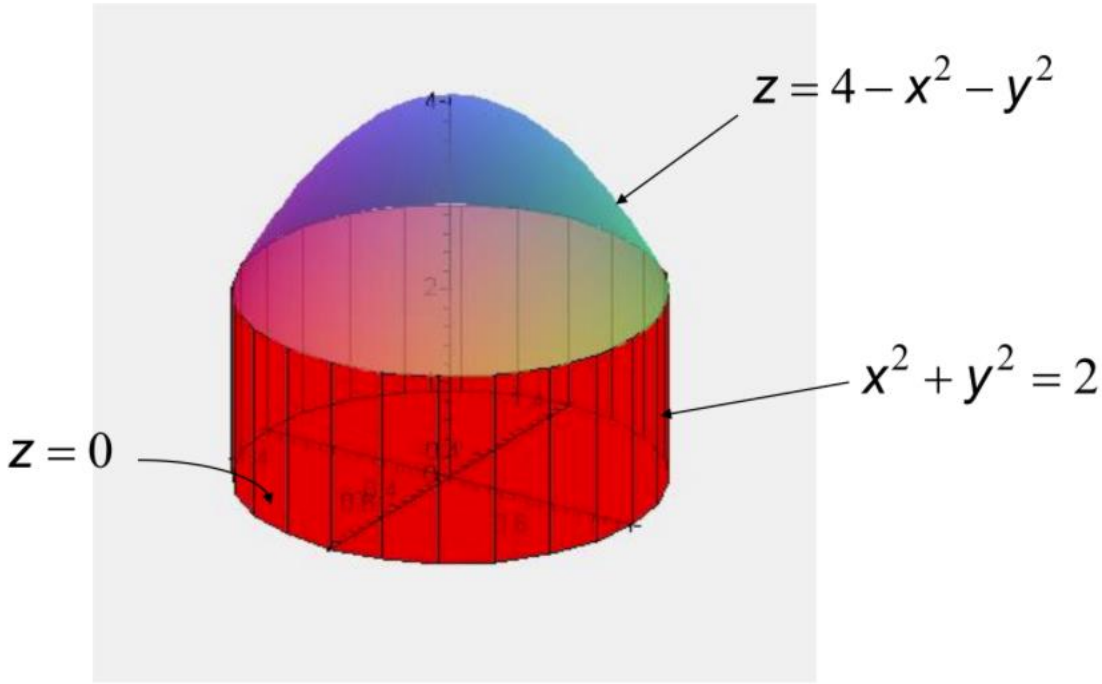


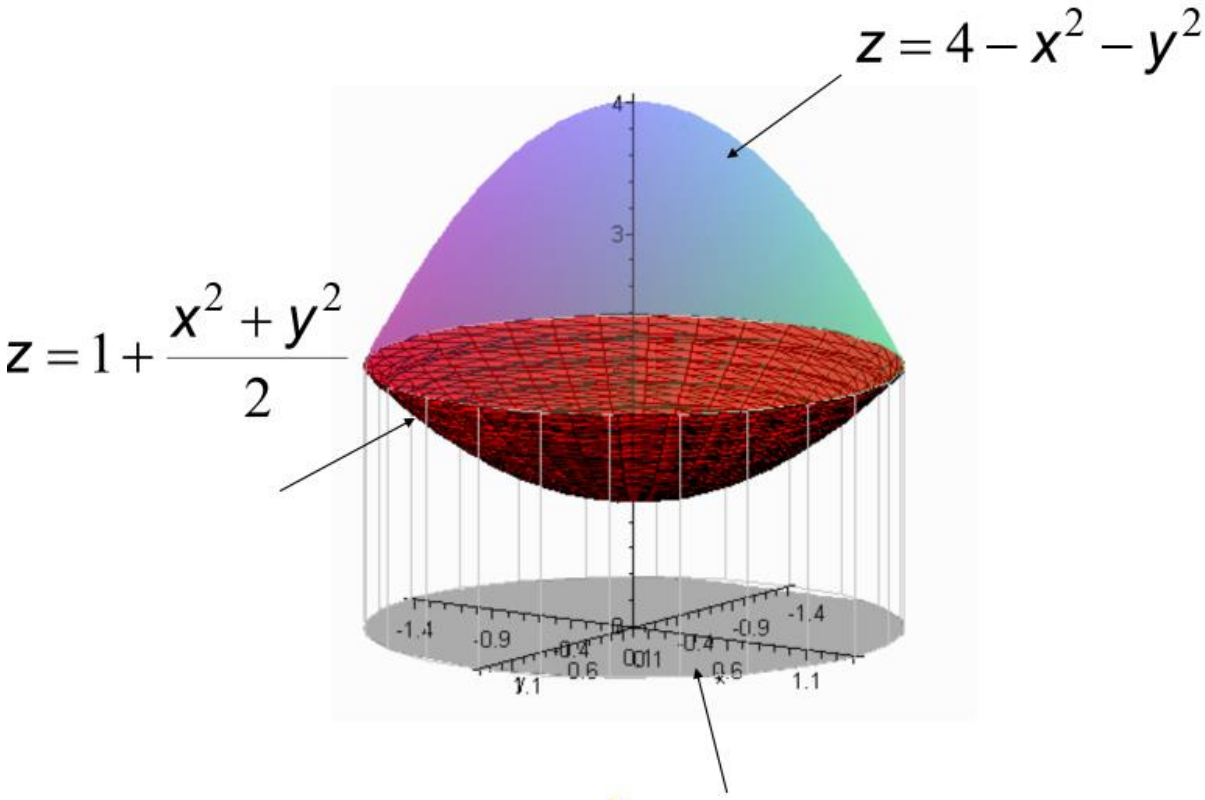
7/ Mặt Hyperboloid hai tầng $\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=-1$

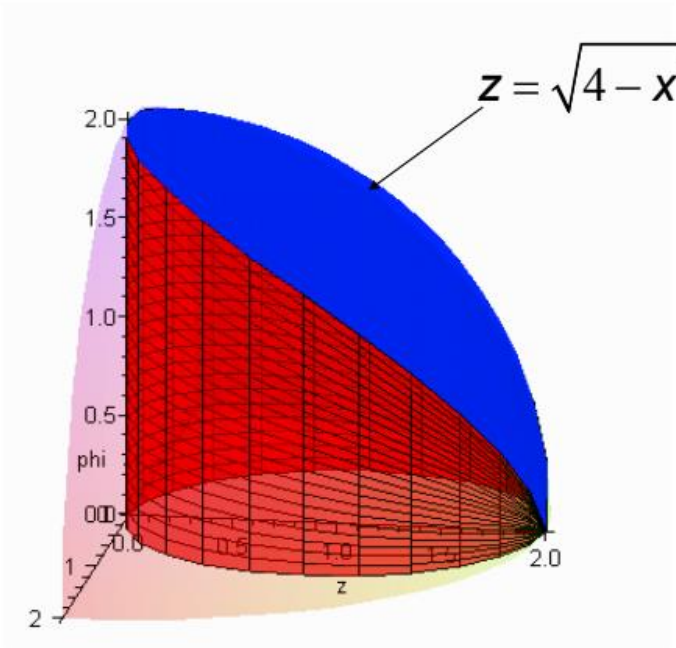


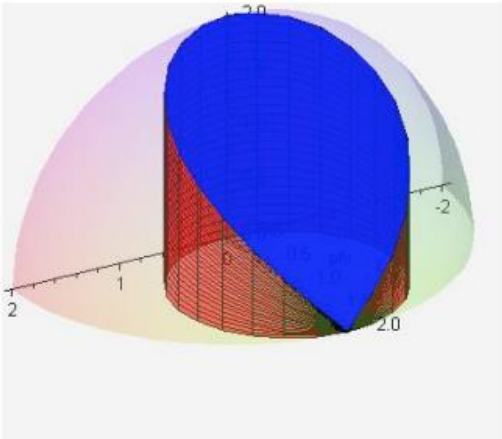


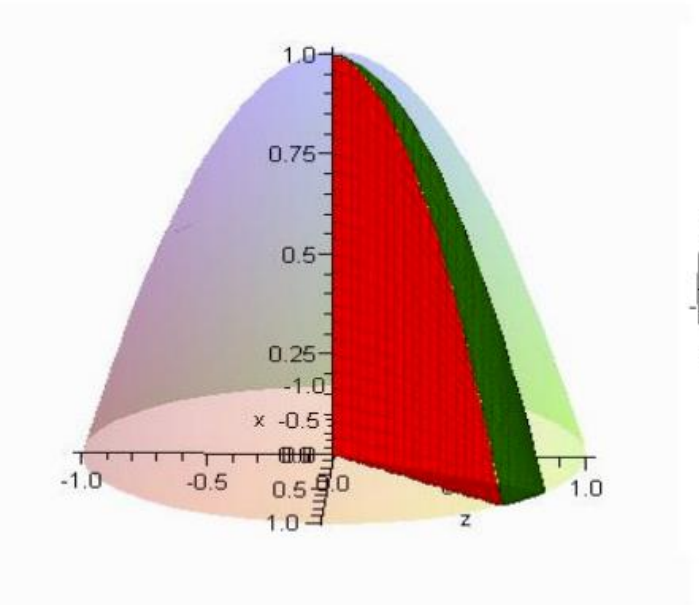
Dùng khi làm bài

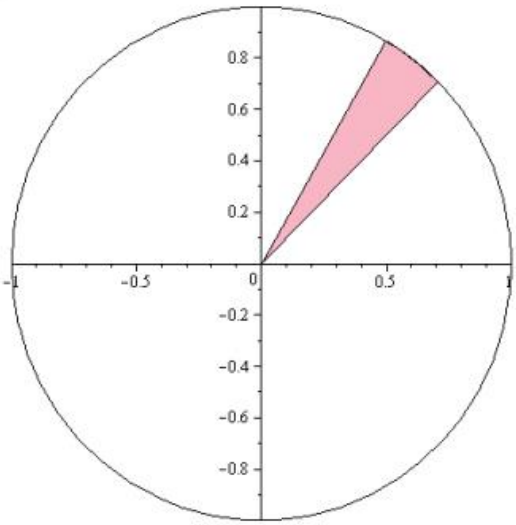

$z = 4 - x^2 - y^2$
$x^2 + y^2 = 2$
$z = 0$

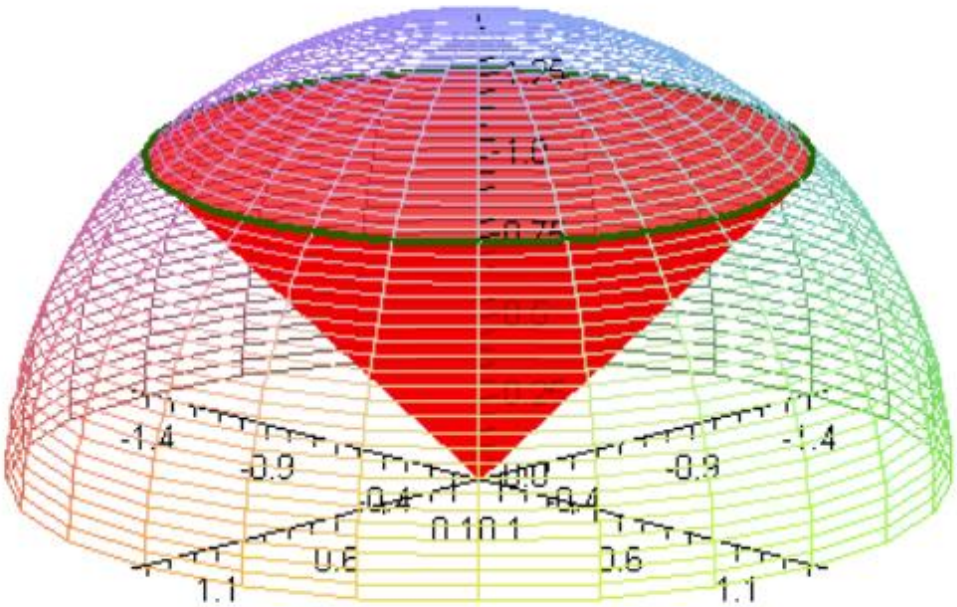

$z = 4 - x^2 - y^2$
$z = 1 + \dfrac{x^2 + y^2}{2}$
Hình chiếu: $x^2 + y^2 \leq 2$


$z = \sqrt{4 - x^2 - y^2}$
2.0
1.5
1.0
0.5
phi
0.0
1
2
z
2.0


-2
1
2
2.0


1.0
0.75
0.5
0.25
-1.0
x -0.5
-1.0
-0.5
0.5
1.0
z


0.8
0.6
0.4
0.2
-1
-0.5
0
0.5
1
-0.2
-0.4
-0.6
-0.8


-1.4
-0.9
-0.4
0.1
0.6
1.1
-1.4
-0.9
-0.4
0.1
0.6
1.1